Early Visual Areas in the Mental Imagery Debate

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Introduction

Mental imagery is a controversial topic in cognitive sciences. At its core, the problem already arises how to define and understand it. According to the most common description, mental imagery is a mental representation in the absence of external stimuli. That is, mental imagery is contrasted with perception where external stimuli reach our senses. To use the anecdotal expression, it is “seeing with the mind’s eye” (Kosslyn, Ganis, & Thompson, 2001). Such a traditional definition implies an analogy between (visual) mental imagery and (visual) perception when it comes to how the experience might be represented in one’s mind. What is at stake here is whether the form of mental imagery can resemble the form of perception.

This question has since become known as the mental imagery debate where two opposing views - the pictorial and the propositional - argue for their own interpretation of the format (but not the content) of mental imagery (Pearson & Kosslyn, 2015). On the one hand, advocates of the pictorial view claim that visual mental imagery is depictive, according to which the structure of the representation is preserved and resembles the original stimulus. For example, a visual mental image of an apple “looks like” an actual apple; in the imaginal space it has the same size, same contours, same colour and so on, as in the actual, perceptual space. Pearson, Naselaris, Holmes and Kosslyn also call it a weak form of perception (2015). That is, if there exists a range of stimulus strength, perception is the strongest “representation” whereas mental imagery is a degraded version of it. As per the authors, this would entail similar visual features to mental imagery as in perception.

Claims on representational similarities might indicate similarities on the level of neurobiology as well. The primary visual cortex (V1) is the first region in the cortex to receive inputs through nerve fibres originating in the eyes (Gazzaniga, Ivry, & Mangun, 2019). Neurons in this region have spatially organised receptive fields; different neurons are tuned to different spatial locations in the perceptual space. Thus, neural representation is topographically organised in this early visual area, which is referred to as a retinotopic map (Gazzaniga et al., 2019). Furthermore, neuroscientific evidence shows that as visual input is processed further towards ventral areas of the occipital cortex, neurons coding for visual information become more and more complex. For example, while V1 codes for edges, V4 neurons become more responsive to shapes (Gazzaniga et al., 2019).

According to the pictorial view, this retinotopic organisation and stream of visual processing is parallel to how visual mental imagery might operate (Kosslyn et al., 2001). The argument goes as follows: if activation of these early visual areas occurs during visual mental imagery, this could support the idea that visual mental imagery is depictive. For example, a less detailed visual mental image might correlate better with V1 than a more detailed representation, which in turn would correlate better with e.g. V3. As such, brain imaging techniques (e.g. fMRI) become crucial in the mental imagery debate.

On the other hand, the propositional view claims that the way visual mental imagery represents information is propositional, i.e. visual mental images represent some object in the same way that a sentence in natural language represents an object. According to Pylyshyn (1981), the information in mental imagery should be understood as symbolic description, rather than a depiction, of the represented object. In this way, a mental image of an apple does not “look like” an actual apple but describes the apple in a symbolic format using many different representations. The meaning of the mental image is therefore determined by its constituent representations, such as beliefs about the world and its objects. What Pylyshyn (1981) calls tacit knowledge about the world partly determines what someone experiences when they mentally visualise something. For example, the reason it takes more time to imagine rotating an object two times, as opposed to one, (Shepard & Metzler, 1971) is not because there is an imaginal space where the rotation of the object is constrained by the extension of the picture-like image as the pictorial view claims, but simply because tacit knowledge about the world states it takes additional time, thus causing the time-increase in the imagining process.

Concerning neuroscientific evidence, Pylyshyn (2002) claims that the format of mental imagery and the involvement of the visual system are independent questions. Nevertheless, Pylyshyn (2002) contends that most neuroscientific evidence does not show the right kind of activation in the visual system during visual mental imagery, nor would evidence of the involvement of the topographically organised V1 provide support for the pictorial view in and of itself. Pylyshyn firstly claims that most neuroimaging studies only report activation in the later, not early, visual cortex. Furthermore, clinical cases of brain damage bring further support against the involvement of the topographically organised areas of the visual cortex. Secondly, in order for activity of the early visual cortex to provide support for the pictorial view, the studies displaying its involvement need to show how the information is mapped topographically in the visual cortex so as to mirror the spatial properties of the mental image, not just that the early visual cortex is active during mental imagery.

The question thus arises: can brain imaging studies settle between the two theories of the mental imagery debate, when it comes to focusing our attention to early visual areas? Would it be a "good enough" evidence for the pictorialists to find activation in early visual areas in visual mental imagery, akin to visual perception? Or would the lack of such activation mean evidence against them, favouring rather the propositionalists? To investigate such questions, this essay will first enumerate brain imaging studies that highlight the role of the visual cortex in mental imagery. This will then be followed by a section describing brain imaging studies that did not find such brain activations. The last section will further elaborate on topics that are indirectly related to the topic, e.g. the role of further brain regions, alternative theories, and possible avenues for future studies.

Neuroscientific Support For the Pictorial View

The earliest study claiming to have found support for the pictorial view was by Kosslyn et al. in 1993. Three separate studies were reported; in the first two, participants were asked to either form mental images of letters in a visual grid or to just look at a grid with a letter presented in it and decide whether an appearing X mark is part of the (imagined or perceived) letter. The second study employed the same procedure but with visually degraded stimuli. In the third study participants were asked to imagine the letters of the alphabet (with closed eyes), either in small or large sizes. Results of the studies showed an overall pattern; the larger a stimulus was imagined, the more anterior of the earlier visual areas (or Brodmann area 17) were activated. In contrast, smaller imagined letters were associated with more posterior parts of this brain region. The authors thus concluded that mental imagery relies on topographically organised early visual areas, similar to visual perception.

While Kosslyn et al. (1993) used PET scans to measure the brain's energy metabolism, fMRI techniques had soon become more prominent in neuroscientific research[1] that rely on oxygenated blood flow in the brain (Gazzaniga et al., 2019). Klein et al. (2000) for example investigated event-related activations of V1 (or the area surrounding the calcarine sulcus, or calcarine cortex) during mental imagery processes. Two events were measured in fMRI; first, participants were asked to generate a mental image (with closed eyes) of an animal, then once hearing either a concrete or an abstract attribute, they were to evaluate whether it is true or not to that animal. Results showed that activation of the calcarine cortex occurred independent of whether a mental image was concrete or abstract. Furthermore, the evaluation phase involved even more activity in this brain region, compared to the image generation phase. The authors contend that such findings show that visual mental imagery involves similar processes to visual perception.

While mentally imagining an animal requires focusing on details of an object, spatial judgments between different structures might involve disparate cognitive processes. Kosslyn et al. (2001) highlight the importance of distinguishing object versus spatial imagery processes.

1 The remaining studies of this section all employed fMRI.

According to the authors, early visual areas are involved only in object imagery but not in spatial imagery. For instance, Ganis, Thompson and Kosslyn (2004) only reported a partial overlap between visual perception and visual mental imagery in the calcarine cortex, but not the rest of the occipital cortex. Although their stimuli involved mentally imagining objects, participants were subsequently required to make spatial judgments (e.g. comparing parts or sizes within the imagined object). That is, it can be argued that while this procedure involved an implicit distinction between object and spatial imagery, the study's results do not necessarily offer support towards the claim made by Kosslyn et al. (2001). Furthermore, Slotnick, Thompson and Kosslyn (2005) used stimuli where spatial geometry was crucial in mental imagery. Participants were presented by rotating arcs (only the outermost part of the stimuli originating from the perception condition) and were asked to "fill in" the object in their "mind's eye". By random time periods a brief red square was presented on the screen, and participants were asked to judge whether this target was inside or outside the imagined object. Results indicated that retinotopic organisation occurred not just in V1 during the visual mental imagery condition, but also in V2 or V3. It is emphasised however that these activations are less strong than those found in the perceptual condition. This would again suggest that defining a clear border between object and spatial imagery is more problematic than it was originally suggested by Kosslyn et al. (2001).

As Ganis et al. (2004) permits, since activation of early visual areas was less emphasised in mental imagery compared to perception, one cannot make clear borders between visual areas (from V1 to V3 e.g.) based solely on visual mental imagery. This further implies that anatomical differences of early visual areas exist between individuals. Bergmann, Genc, Kohler, Singer, and Pearson (2016) investigated such individual differences with the binocular rivalry paradigm. Participants were first cued which coloured gratings to imagine (red or green Gabor patches), followed by seven seconds of imagination phase. After rating the vividness of their experience, a brief stimulus was flashed to the participant's eyes; crucial to the paradigm, one Gabor patch (red vertical) was flashed to one eye and the other (green horizontal) was flashed to the other eye. Participants were then asked to indicate which coloured grating they had seen; imagery strength can be calculated from matching imagery and perceptual stimuli. Two further conditions varied either the orientation or the location of the Gabor patches, thus offering a subsequent variable to measure: imagery precision. This paradigm is considered an objective measurement of mental imagery as it can quantify an imagined percept's priming effect on subsequent perception (Bergmann et al., 2016). Results indicated that participants having smaller V1 were more likely to have stronger mental imagery. While V2 and V3 were also mapped, these regions did not reach statistical significance for imagery strength. A second finding was that the larger V1 participants had, the higher their imagery precision was. The authors concluded that anatomical differences in V1 have to be considered not just in perception but also in mental imagery.

The empirical evidence so far presented builds on the linear fashion of a typical study design: participants follow a procedure of mental imagery, while imaging techniques visualise their brain activity. Machine learning, however, offers the opportunity to turn these steps backwards; through brain activation images an algorithm can detect for example the contents of a mental image. Thirion et al. (2006) for instance used two algorithms (successfully) on retinotopic maps of V1 to classify the contents of mental images participants had earlier imagined in scanning sessions. Naselaris, Olman, Stansbury, Ugurbil, and Gallant (2015) also showed that low-level visual features (such as space or orientation) can be encoded from mental images, similar to perception. Such novel techniques thus can be seen as converging evidence towards the view that mental images might be pictorial in their nature.

Neuroscientific Support Against the Pictorial View

The propositional argument revolves more around behavioural data and theoretical discussion than the pictorial view. As Pylyshyn (2002) emphasises, he believes that the issue at hand when it comes to the propositional-pictorial disagreement is separate from questions of the involvement of the visual system. However, as many authors believe the argument for the pictorial view from the involvement of early visual areas in visual mental imagery to be quite striking, a lack of such an involvement should provide an apt counter argument. In that case, what is the neuroscientific evidence against activity in the early visual areas in visual mental imagery?

A study by D'Esposito et al. (1997) used fMRI to investigate the involved brain regions in visual mental imagery. The participants were aurally presented with a random noun, where the referents were judged either difficult or easy to imagine based on being abstract (e.g. *justice*, *democracy*, etc.) or concrete (e.g. *chair*, *house*, etc.) respectively. After being presented with the word, the subjects were to mentally visualise the referent of the word when being concrete, or passively listen if it was abstract. The brain regions that revealed more activation during the concrete condition than the abstract condition were considered to be involved in visual mental imagery, which most reliably included areas in the occipito-temporal junction, including the fusiform gyrus. The involvement of these regions therefore indicates that the visual association cortex is most prominently involved with visual mental imagery. The results showed no activation in the primary visual cortex.

Three years later, Knauff et al. (2000) conducted an experiment using fMRI. The participants were presented with a geometrical figure on a grid, where the task consisted of judging whether a highlighted square in the grid fell within the bounds of the figure (the perception condition), which is then repeated when the figure was removed (the mental imagery condition). The authors primary goal was to see whether the primary visual cortex and nearby areas are involved in visual mental imagery; whereby activation in V1 should be seen during the imagery condition, not just during the perception condition. No support for the involvement of V1 was observed, where the data only showed weak activations during the imagery condition in V1 for two of the 10 participants. Therefore, Knauff et al. conclude that the primary visual cortex is not necessary for visual mental imagery. They also stress the central role of the higher visual areas and the occipital-parietal pathway, which are known for their incorporation of cognitive and visual functions.

As Fulford et al. (2018) observe, the higher visual association areas are positively correlated with increased vividness of imagery, including the precuneus, posterior cingulate and middle temporal lobe; while they note a negative correlation for the primary visual cortex - although, the areas most positively correlated with low vividness lie mostly in the frontal lobe and auditory cortices. In combination with their fMRI experiment, they also note a trend of a negative association of the early visual cortex and a general variation of its contribution with visual imagery in their literary review - sometimes showing a deactivation of the early visual areas during mental imagery. They remark that an explanation for this variation is unclear, but higher visual areas are consistently associated with increased vividness in previous studies as well.

To get a clearer overview of the neuroscientific evidence on the involved brain areas, Spagna et al. (2021) conducted a meta-analysis of 46 fMRI studies of visual mental imagery conducted over the past 30 years. They noted that the evidence in previous studies has been, at best, ambiguous when it comes to the involvement of early visual areas. However, on account of their analysis, it becomes clear that the evidence lacks support of its involvement since the meta-analysis of brain activity during visual mental imagery shows no consistent activation in the early visual areas. Spagna et al. suggest an alternative model of visual mental imagery, contrary to the standard pictorial model, based on the regions which showed consistent activity (which will be considered with more detail in the discussion). They additionally stress that the evidence from their meta-analysis is very much in line with studies on the effect of brain lesions on mental imagery. In fact, patients with brain damage isolated on the occipital lobe are documented to still exhibit unimpaired visual mental imagery, concordant to the results of their meta-analysis.

One such study was conducted by Gelder et al. (2015), where a patient with bilateral lesions on the early visual areas, resulting in complete cortical blindness, was asked to perform visual mental imagery. The fact that the activated regions of the patient's brain overlapped with the activation in the control group, in combination with a subjective report of the mental imagery, suggests that the patient successfully performed visual mental imagery. This implies a dissociation between perceptual and imagery deficits. Not surprisingly then, Dulin et al. (2008) direct the attention to a double dissociation among several studies of deficits among patients with neurovisual disorders following cortical damage. Studies exhibit patients without occipital damage that display mental imagery impairments, while patients with occipital damage display no imagery impairment (e.g. Bartolomeo, 2002). This runs contrary to what the pictorial view would expect: if visual mental imagery depends on a pictorial display from the topographically organised early visual areas, then imagery deficits should result when those areas cease to function.

Discussion

How mental imagery "is" for a person is a question concerning philosophers and scientists for many years now. The mental imagery debate sparked new flames; while the pictorialist view contends that visual mental imagery should be understood in terms of visual perception and depiction, the propositionalist view asserts it to have a symbolic structure and descriptive representation. The former party further argues that if early visual areas of the occipital cortex have a similar activation in visual mental imagery as in visual perception, this alone shows support for a depictive explanation. The latter party finds this argument weak, opposing the necessity of early visual areas in mental imagery. While Pearson and Kosslyn (2015) claim that the debate has been ended due to recent neuroscientific evidence, our paper displays an opposite *interim* conclusion; interim, by which we mean that the debate is far from over. For the pictorial claim, we lined up studies that showed increased activity in V1 (Kosslyn et al., 1993; Klein et al., 2000); presented supporting evidence in a separate but parallel debate on object versus spatial imagery and the involvement of early visual areas such as V1, V2, and V3 (Ganis et al., 2004; Slotnick et al., 2005); highlighted the necessity to consider individual anatomical differences (Bergmann et al., 2016); and peaked into machine learning techniques (Thirion et al., 2006; Naselaris et al., 2015). For the propositional claim, we enumerated studies that presented no increased activity in V1 (D'Esposito et al., 1997; Knauff et al., 2000); enlisted a literature review (including an fMRI experiment) and a meta-analysis stating a lack of evidence of early visual areas in visual mental imagery (Fulford et al., 2018; Spagna et al., 2021); and involved further brain lesion studies where although the occipital lobes were damaged patients still retained visual mental imagery (Dulin et al., 2008; Gelder et al., 2015). Based on these findings, the main research question of our paper cannot be answered with confidence. How mental imagery "is" (or rather, how the format of mental imagery is) for a person cannot be decided based solely on early visual areas.

This constraint would be consolidated by expanding the discussion to other involved structures. As Kosslyn et al. (2001) note, the activated areas differ according to what the participant imagines, even within visual mental imagery. The parahippocampal area activates while visualising spatial layouts of scenes (like that of a room in a house); which contrasts to the visualisation of faces, which activates the fusiform face area. This is also seen in the previously mentioned difference in activation between spatial and object imagery. As claimed by Ganis et al. (2004), the early visual areas show greater activation when visualising details in an image, while visualising spatial attributes activates the posterior parietal cortex and not the medial occipital cortex. The activated areas also differ when contrasting visual and auditory mental imagery: auditory mental imagery activates areas involved in auditory perception (e.g. bilateral associative auditory cortex) - although, not the primary auditory cortex. The additional brain regions identified in visual mental imagery by Spagna et al. (2021) included bilateral areas in the fronto-parietal networks and anterior insular cortices; while unilateral activations were revealed in the left fusiform and parahippocampal gyri, specifically the right precuneus and left anterior calcarine sulcus. Based on these activations and a lack of activation in the early visual areas, they suggest a new model of visual mental imagery, which claims that the core of the network, that they label the Fusiform Imagery Node (FIN), is located in the left middle fusiform. The FIN is provided semantic information from the left anterior temporal lobe. This process is initiated by prefrontal regions, supported by evidence which indicate that visual mental imagery requires the activation of fronto-parietal networks that also support higher level cognitive processes such as top-down spatial attention. These prefrontal networks also maintain the image in order to complete certain tasks.

While several other areas in the brain above the purported involvement of the early visual areas are certainly involved, it is important to emphasise a second caveat of the mental imagery debate. The two opposing theoretical views might not cover the full spectrum of mental imagery. For example, a third hypothesis derives from embodied cognition. Palmiero et al. (2019) argue that how mental imagery is represented is less important than how we use our bodies in the world and how enacting through motor procedures underlies our mental experiences. While the embodied approach has several branches (e.g. simulation, enactive, or sensorimotor), common to them is an overlap of brain regions between sensory and motor areas. This would then indicate a combination of what a person internally represents and how the person interacts with the outside world. As such, the traditional mental imagery debate might need a revision, and future studies should not limit their scope to only the two opposing sides.

Taking one side in the mental imagery debate might come naturally, almost intuitive. For it can always be the case that we arrive at the conclusion that our mental imagery is either pictorial or propositional (or something else) based on personal reflection. Reisberg, Pearson and Kosslyn (2003) point our attention towards this concern where it was found that philosophers' and scientists' intuitions of their own mental experiences shape how they argue about the format of mental imagery. Although one of the co-authors was Kosslyn himself (the originator of the debate), such an investigation represents a good amount of self-reflection. After all, mental imagery is a subjective phenomenon where everybody has the authority to decide for themselves how it manifests in their experience.

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